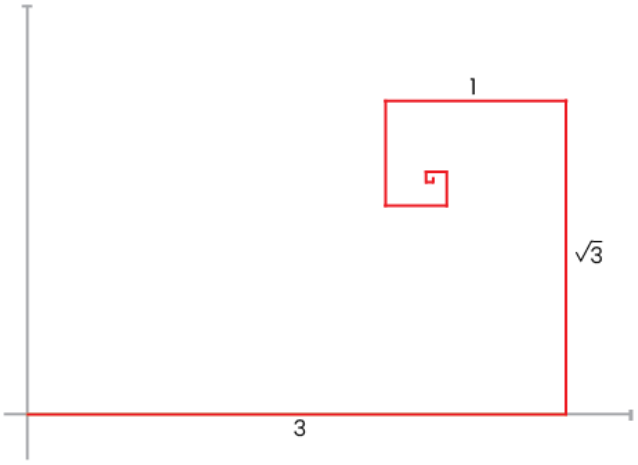
Ritangle 2026

Stage 1 - Question 6

Project π*

October 17, 2026

Topic - Geometric series



*Solved by Nihal Sharma.

1 Problem

The problem statement was as follows:

Unknown

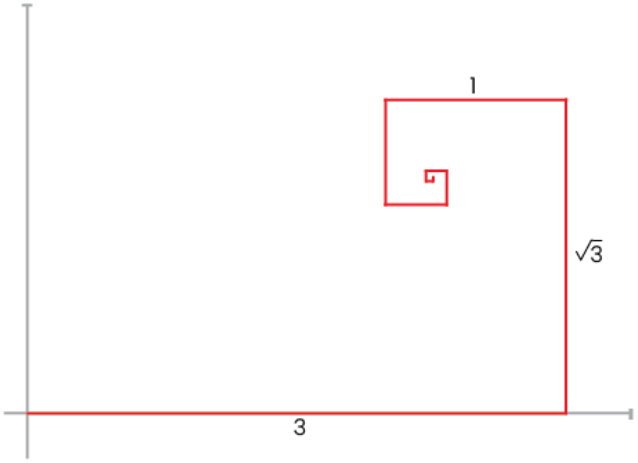


Figure 1: Reference image

I'm assuming the question is to calculate the distance from the origin of the point the infinite spiral converges to

2 Solution

The key insight for this question is to recognise that spiral can be broken down into two geometric series

Let the point the infinite spiral converge on be some point P with co-ordinate (x, y)

The x co-ordinate is the value of an infinite geometric series with first term 3 and common ratio $-1/3$, so $x = \dfrac{3}{1 - -1/3} = 9/4$

The y co-ordinate is the value of another infinite series with first term $\sqrt{3}$ and common ratio $-1/3$, so $y = \dfrac{\sqrt{3}}{1 - -1/3} = \dfrac{3\sqrt{3}}{4}$

From this we can conclude that $|OP| = 3(\sqrt{3^2 + 3})/4 = 3\sqrt{3}/2$

Final Answer Summary:

$$\text{Distance} = \frac{3\sqrt{3}}{2} \implies \textit{Hint: ???}$$